

Project Name: Green Sun

Project Idea:

1- The Target Problem:

High costs of relying on non-renewable resources for electricity generation and the accompanying challenges:

- **Reliance on importing oil and gas**, which puts pressure on the state budget.
- **Instability of these resources due to international circumstances** (such as the war in Ukraine).
- **High maintenance and operating costs** for equipment that relies on fossil fuels.
- **Significant negative environmental impact:** Pollution of air, water, and soil, and the resulting health damages to humans, animals, and plants.

2- The Proposed Solution:

A clean and practical alternative had to be considered.

Solar energy represents the optimal solution for Egypt due to the availability of sunshine throughout the year across most of its lands.

- **The use of solar panels provides a permanent, clean, and environmentally friendly source of energy.**

Team Members:

Somaya Yasser Ali (Leader)

Alaa mahrous Hassan

Christeen Nabil Abdallaha

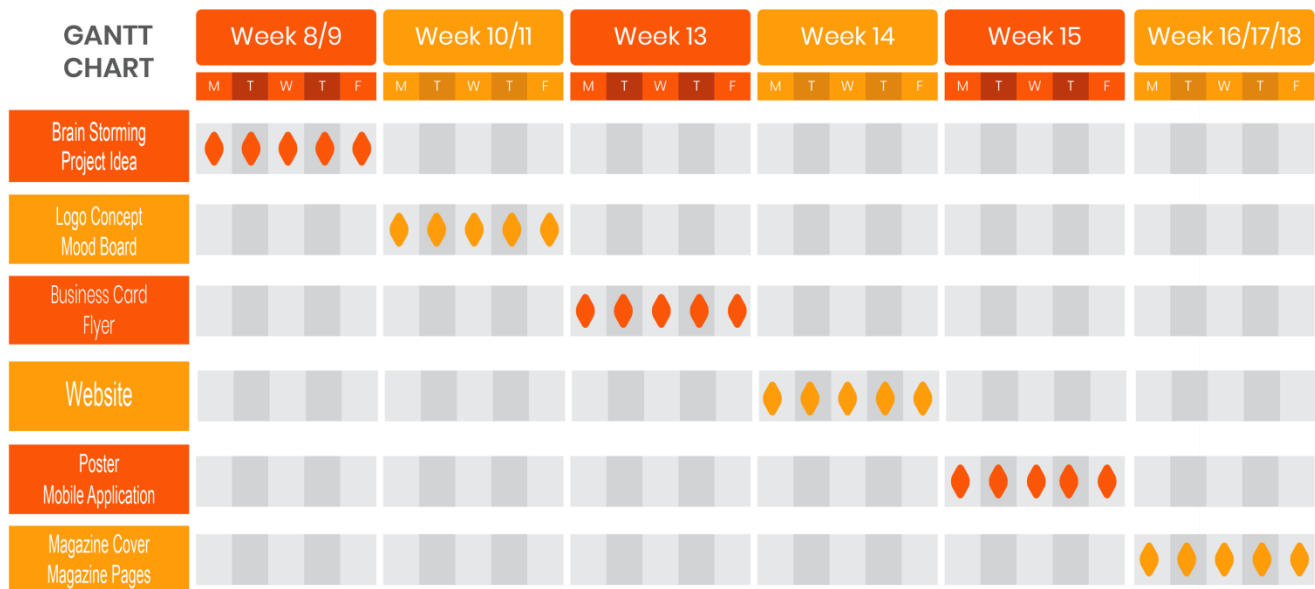
Maram Mohamed Abd Alsatar Ibrahim

Mariam Amr Mostafa

Menna Tarek Nassef

Project Planning & Management.

Project Plan – Timeline (Gantt chart)



Task Assignment & Roles – Defined responsibilities for team members.

Logo & Business Card	Christeen
Flyer	Mariam
Website	Somaya
Mobile Application	Maram
Magazine Pages	Menna
Social Media Ads	Alaa

Work Plan

Research & Analysis

Audience personas



Visual Identity

Logo Design



The logo features a large yellow circle. Inside the circle, the letter 'G' is formed by a warm orange semi-circle with short green lines radiating from its top-left, resembling sun rays. To the right of the 'G', the words 'Green' and 'Sun' are stacked vertically in a bold, green, sans-serif font. In the top-left corner of the slide, there is a small green wavy line. In the bottom-left corner, there is a small orange wavy line.

LOGO IDEA

The logo is inspired by the name "Green Sun," where the letter G forms a half-circle in warm orange, visually representing the rising sun. Soft green rays extend outward to symbolize clean, renewable solar energy.

DESIGN CONCEPT

The design merges two core elements:

- The sun: representing energy, power, and brightness
- The green color: symbolizing sustainability, nature, and a cleaner future

The combination visually reflects the brand's mission: promoting natural, clean, and renewable solar energy solutions.

COLORS

		
#00a651	#f99e1c	#fcf172

TYPOGRAPHY

	
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Trys and skitch

Symbols					
Colors					
Skitch and final logo					
					

Main Designs Business Card

business card



FLYER



OTHER IDEA



POSTER

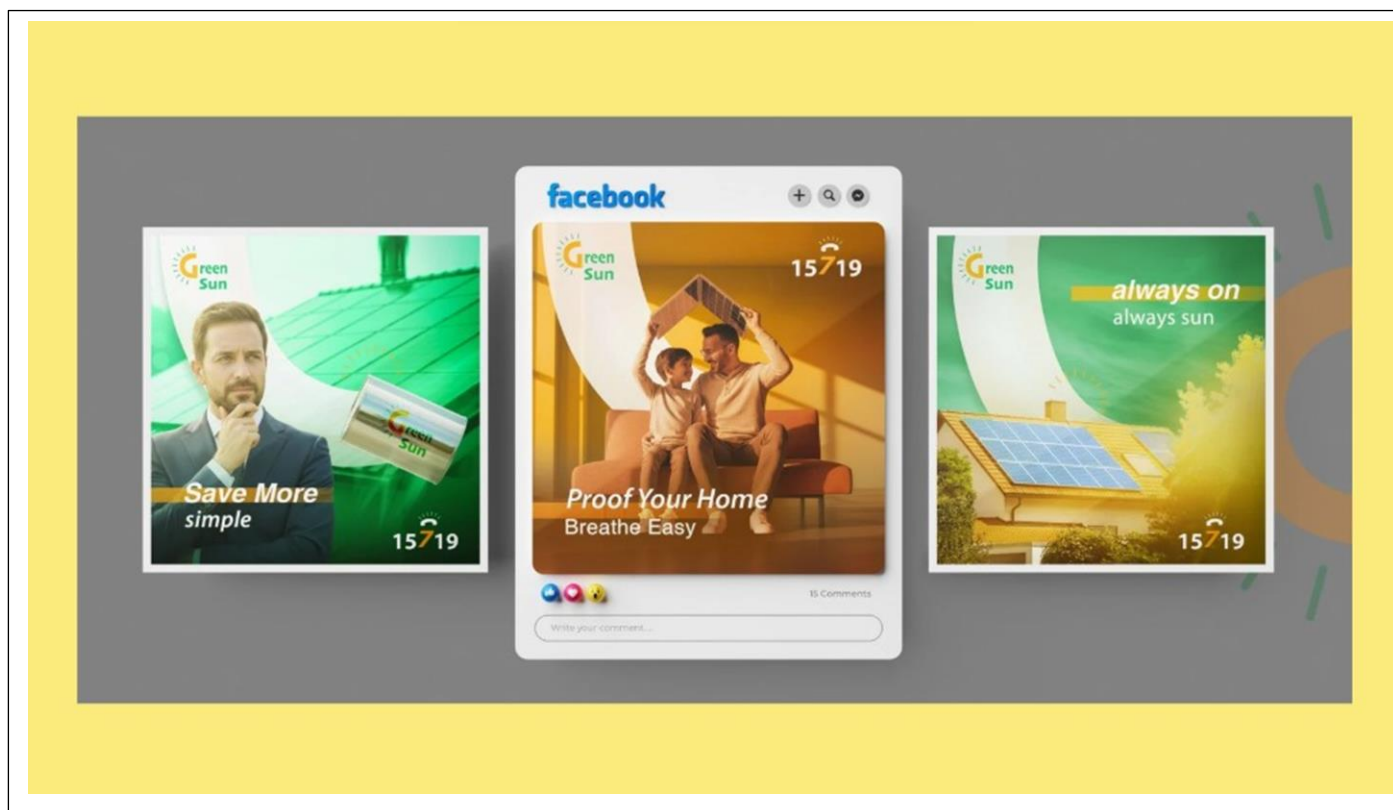
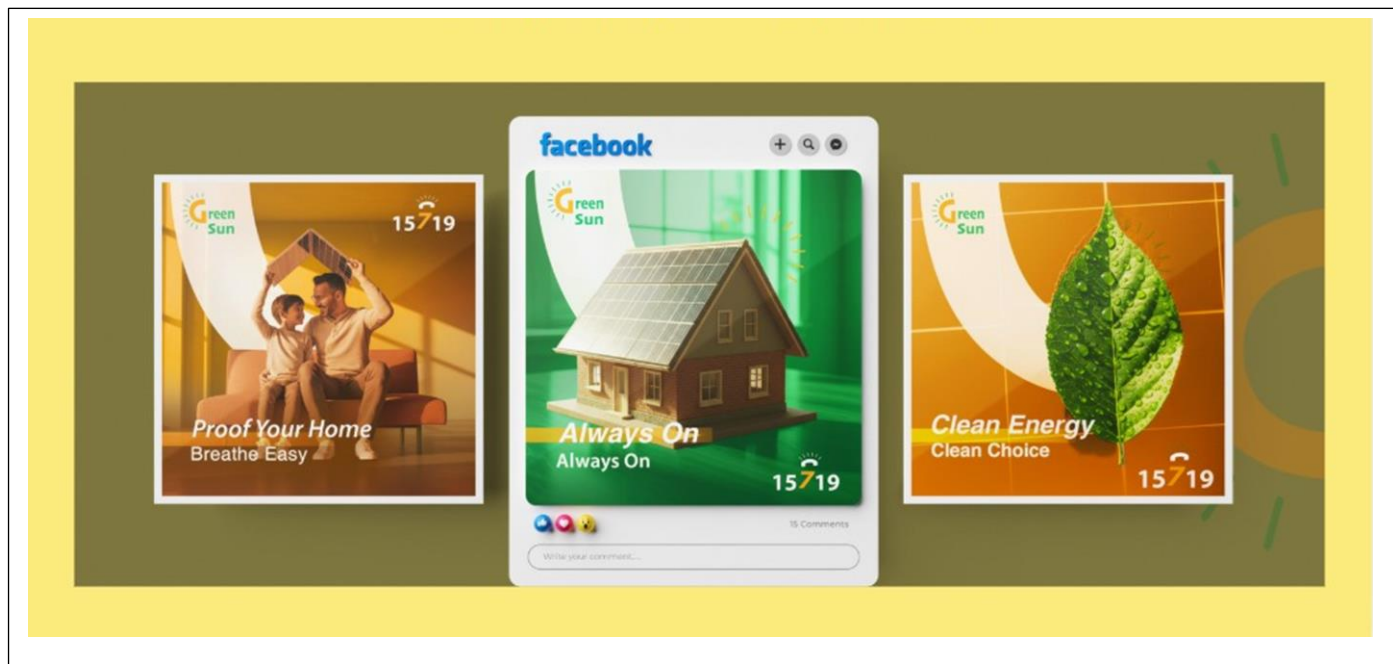


MAGAZINE

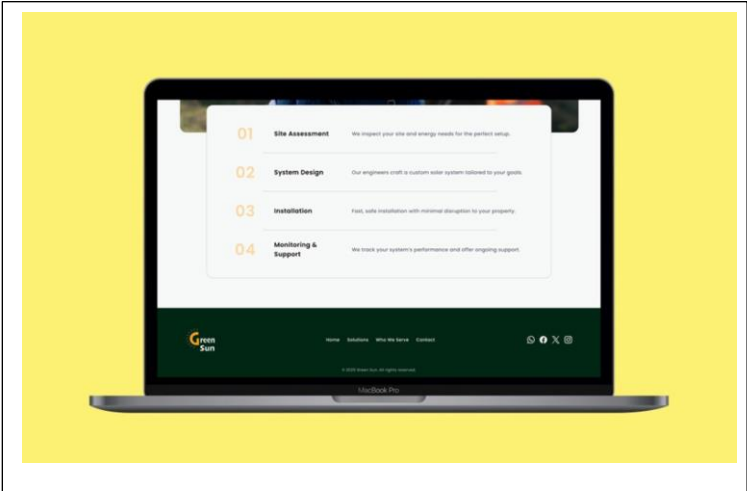
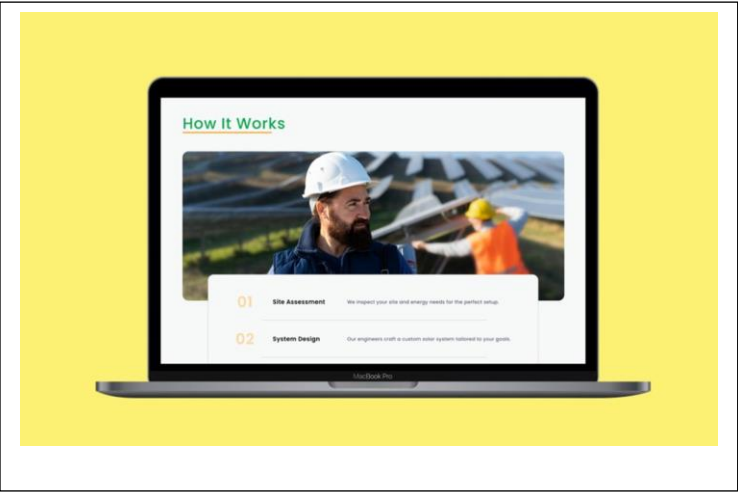
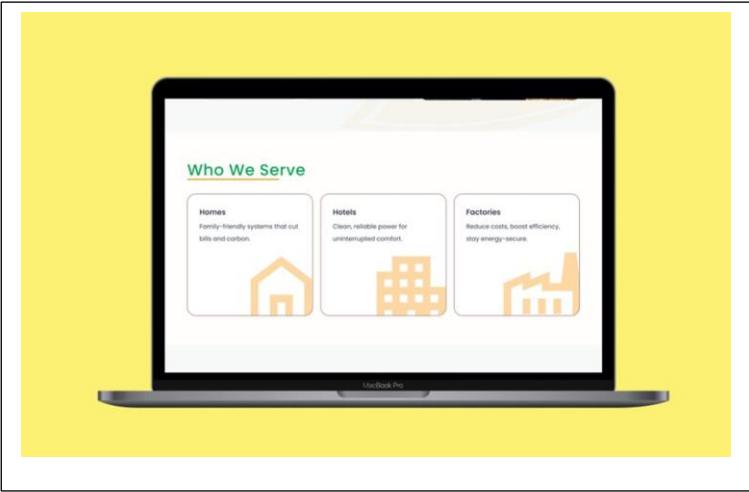
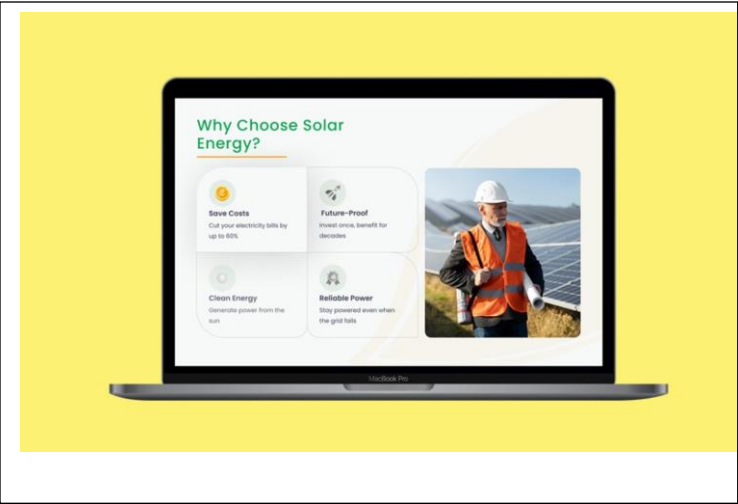




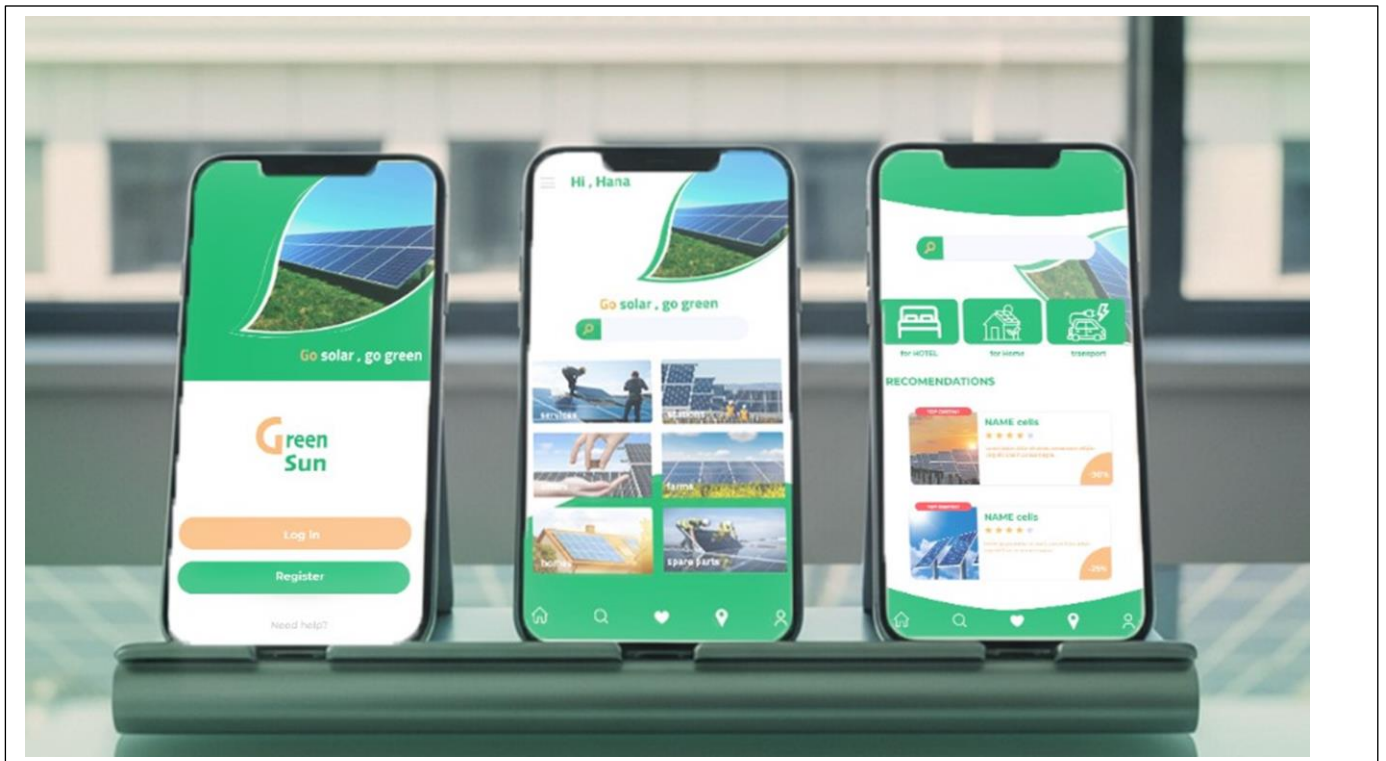
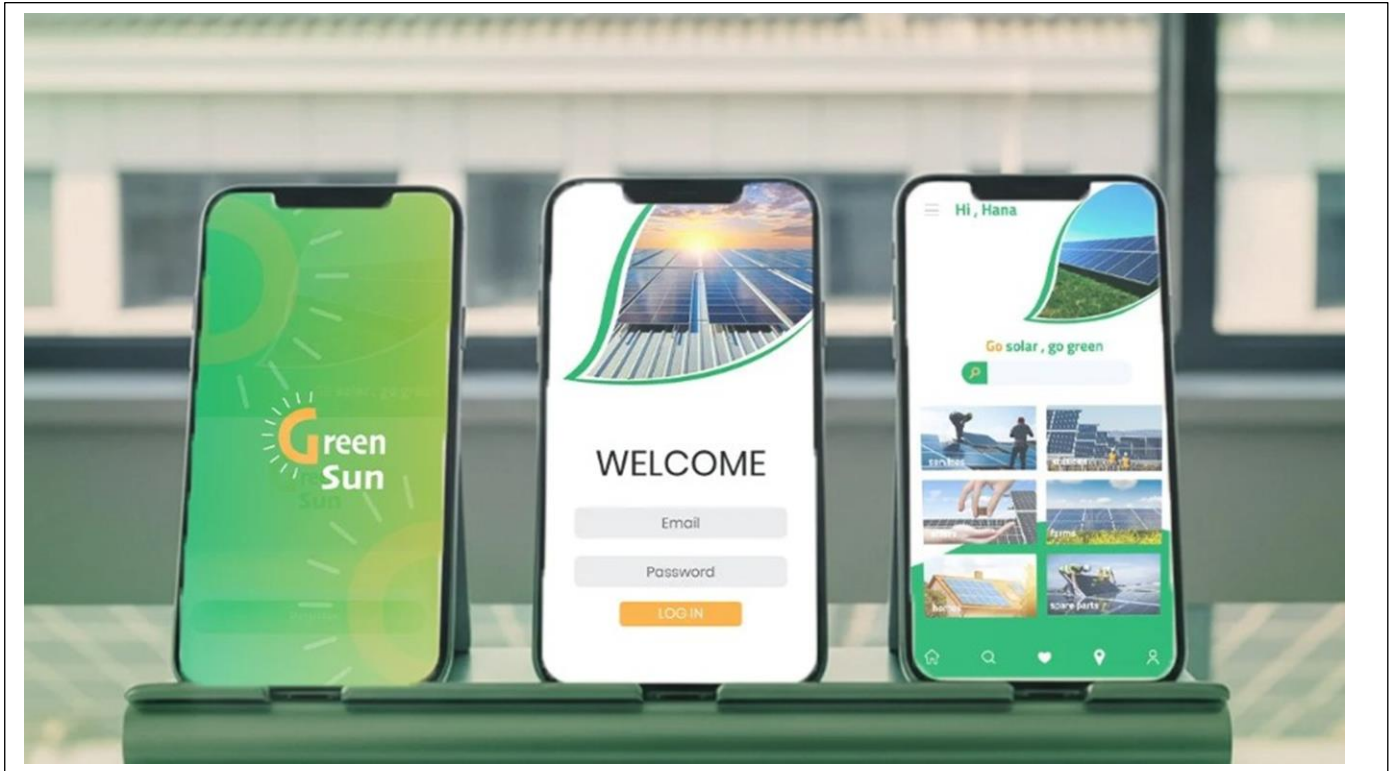
Social Media designs



Website



Application



Complementary products

Mood Board					
Competitors					
Colors					
Fonts type	Serif, Sans-Serif	Handwriting	Sans-Serif	Serif	Sans-Serif
Symbols	 	  	  	 	 

Final Presentation

Link:

KPI

Key Performance Indicators (KPIs) for a Solar Energy Project

To make these KPIs measurable and effective (following the SMART framework), targets are included based on common assumptions for large-scale solar projects in the region (e.g., Egypt).

1. Technical and Production KPIs

KPI	Target Value/Description	Significance
Capacity Factor (CF)	Target: $\geq 22\%$	The actual energy generated compared to the theoretical maximum. This percentage reflects the real output efficiency, factoring in weather and sun availability. Solar projects in Egypt typically range from 20% to 25%.
Performance Ratio (PR)	Target: $\geq 80\%$	Measures the overall quality and efficiency of the PV system, independent of weather conditions. A PR of 80% means only 20% of potential generation was lost to system inefficiencies (inverter, wiring, heat, etc.).
Annual Energy Yield	Target: $\text{1.8 Million kWh / year}$ (Assuming a 1 MW project)	The total amount of electricity actually produced by the solar system during a year.
System Downtime	Target: $\leq 1\%$ of total annual operating hours	The total time the system is not producing due to faults or maintenance. This is a measure of system reliability.

2. Financial and Economic KPIs

KPI	Target Value/Description	Significance
Payback Period	Target: 5 to 7 years	The time required to recover the initial investment cost through savings from avoided electricity purchases. This is a key metric for investor viability.
Net Present Value (NPV)	Target: A large positive value (e.g., \$$\text{EGP } 50,000,000$)\$ over 20 years)	Measures the long-term profitability by calculating the present value of future cash flows minus the initial investment, discounted to today's value.
Levelized Cost of Electricity (LCOE)	Target: Less than utility purchase price (e.g., \$$\text{EGP } 1.2 / \text{kWh}$)\$)	The average net cost per unit of electricity generated by the solar system over its lifetime. The goal is to be lower than the grid electricity price.
Annual Financial Savings	Target: \$$\text{EGP } 3,000,000 / \text{year}$\$)	The projected monetary value of costs avoided due to self-generated solar power.

3. Environmental and Socio-Economic Impact KPIs

KPI	Target Value/Description	Significance
\$$\text{CO}_2$ Emissions Avoided	Target: \$$\text{900 Tons}$\$ of \$$\text{CO}_2$ per year	The amount of carbon dioxide emissions prevented by using solar energy instead of fossil fuels. This is the main positive environmental impact metric.
Number of Benefiting Customers/Subscribers	Target: \$$\text{2,500}$\$ direct households/families	The total count of homes, institutions, or individual clients receiving electricity directly from the solar

KPI	Target Value/Description	Significance
		project (especially relevant for community or off-grid projects).
Jobs Created	Target: $\text{\\$10}$ permanent roles (O&M and management)	The number of direct, long-term employment opportunities generated by the project's operation and maintenance phases.
Reduction in Fossil Fuel Consumption	Target: $\text{\\$500,000}$ Liters of Fuel Oil Equivalent / year	The volume of traditional fuel (e.g., natural gas, oil) saved due to the solar energy production.

Instructor: Dr / Suzan Farahat
project files:

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